tubulin: relative 3Di/AA ratio per branch (rho; colour = log2 rho; outgroup-rooted for display)

FtsZ (35)
FtsZ (4)
euk_FtsZ (4)
FtsZ (1)
euk_FtsZ (4)
FtsZ (5)
euk_FtsZ (14)
FtsZ (11)
FtsZ (33)
Asgard_FtsZ (1)
FtsZ (31)
CetZ (15)
tubulin_divergent (9)
tubulin_delta (9)
tubulin_divergent (3)
tubulin_epsilon (29)
tubulin_epsilon (3)
artubulin (3)
tubulin_gamma (53)
BtubA (1)
BtubB (1)
tubulin_alpha (62)
tubulin_beta (60)

CetZ (0.76)
CetZ+tubulin (0.88)
tubulin_delta (1.13)
tubulin_epsilon (0.45)
tubulin clade (euk+artubulin+BtubAB) (1.21)
artubulins (0.74)
tubulin_gamma (0.89)
tubulin_alpha (0.68)
tubulin_beta (0.89)

log2 rho (3Di/AA relative to tree)
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0

UFBoot<95 or b_AA<0.02
b_AA>2 (saturated, excluded)
robust high flag
robust low flag
Eukaryota
Asgard
Archaea
Bacteria

0
1
2
3
4
5
6
7
8
AA substitutions/site (IQ-TREE LG+G4, fixed ML topology, MAFFT columns)